



Design of Slot Antenna with AMC for Wideband Applications

ABSTRACT

The rapid development of wireless communication systems has created a growing demand for antennas that can provide high bandwidth, high gain, and reliable radiation performance while maintaining a compact structure. Conventional microstrip and slot antennas are widely used because of their low profile, simple fabrication, and low cost. However, they often suffer from limited bandwidth, lower gain, and unwanted radiation in certain applications. This project focuses on the design of a Slot Antenna with an Artificial Magnetic Conductor (AMC) structure for wideband applications to overcome these limitations.

The proposed antenna combines a slot antenna with an AMC surface to improve its overall electromagnetic performance. The slot antenna is designed to operate over the required frequency range, while the AMC structure is placed beneath or around the antenna to control the propagation of electromagnetic waves and reduce backward radiation. The AMC surface provides an in-phase reflection, which helps improve the constructive interaction between the direct and reflected waves. As a result, the antenna can achieve enhanced impedance bandwidth, higher gain, improved radiation efficiency, and a more directional radiation pattern.

The expected outcome of this project is a compact and efficient slot antenna with improved wideband performance compared with a conventional slot antenna. The proposed design can be useful for modern wireless communication applications such as 5G communication, WLAN, satellite communication, and other high-frequency wireless systems. Overall, the project demonstrates how AMC technology can be effectively integrated with a slot antenna to achieve wide bandwidth, high gain, and a unidirectional radiation pattern.

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